

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th December 2018

Subject CT1 – Financial Mathematics

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)**
- i)** State four uncertainties attached with the rate of return earned on the fixed interest Government bonds and give your comments on them. (4)
 - ii)** Discuss the risks involved in the swap agreement for the investor? (2)
 - iii)** Person A bought a European call option from B and sold a European put option to C against the same underlying security with the same strike price and term. Explain who should exercise the option in the given scenarios and why:
 - a) Share price on the exercise date is greater than the strike price. (1.5)
 - b) Share price on the exercise date is lesser than the strike price. (1.5)
 - iv)** A fixed interest security pays coupons of 7% per annum (p.a.) payable half-yearly in arrear and is redeemable at 114%. Five months before the next coupon is due, an investor negotiates a forward contract in which he agrees to buy INR 1,00,000 nominal of the security in 8 months' time.

The current price of the stock is INR 78.94 per INR 100 nominal and the risk-free force of interest is 3.5% p.a. Calculate the forward price. (4)
- [13]**
- Q. 2)**
- i)** Explain "equation of value". (1)
 - ii)** A Government issues a 91 day treasury bill at a simple rate of discount 7% per annum. Calculate the rate of return per annum convertible half yearly received by an investor who purchases the Bill and holds it to maturity. (3)
 - iii)** Yash wishes to go on a dream vacation in 10 years' time. He saves a portion of his monthly salary and invests it in a recurring deposit which earns a return of 8.15% per annum effective. Expected inflation over this period of 10 years is 5% per annum effective.

How much should Yash save every month so that he gets INR 50 lakhs in real value term (i.e. as per today's monetary value) at the end of 10 years? Assume his next monthly salary is due exactly after one month from now. (3)
- [7]**
- Q. 3)**
- i)** Explain the notation $a^{(p)}_{n|}$ (1)
 - ii)** Prove algebraically that $a^{(p)}_{n|} = i/i^{(p)} a_{n|}$. Also explain the above equation with general reasoning. (4)
 - iii)** Calculate $(I\ddot{s})_{\overline{10}|}$ @ 11% per annum effective **and explain** the notation. (3)
 - iv)** An investor has an amount of INR 10,000 to invest in two investment options. She has to put at least 25% of the capital in each of the options. Calculate the maximum accumulated value of the investments after a period of 5 years.

Under each of the options, the investment made at time 0 accumulates at:

Option A: Compound discount rate of 7% per annum convertible half-yearly for 2 years followed by nominal rate of interest of 8% per annum convertible quarterly for the balance period.

Option B: Force of interest $\delta(t) = 0.05 + 0.003t$ for $0 \leq t \leq 5$. (4)
[12]

Q. 4) A student has taken educational loan of INR 50 lakh for higher studies. The loan amount will be disbursed in four equal half yearly installments (i.e. over a period of 2 years) payable in advance. The loan is repaid over a period of 10 years in equal monthly installments. First repayment installment will be paid at the end of 25th month from the first date of loan disbursement.

- i) Calculate the amount of each monthly installment assuming a rate of interest 10% p.a. effective. Assume interest is charged on the disbursed loan amount from the date of such disbursement. (4)
 - ii) Calculate the interest component (in amount) of first monthly installment. (2)
 - iii) After 5 years of repayment, the student wishes to enhance monthly repayment installment so that the remaining loan is repaid over next 2 years. Calculate the new monthly installment amount assuming the same effective rate of interest as above. (5)
 - iv) Calculate the amount of interest paid during last 2 years (after loan reschedule (as in (iii) above). (2)
- [13]

Q. 5) A builder is constructing a shopping mall which will have 42 shops. The total cost of construction is expected to be INR 105 crores out of which INR 10 crores will be incurred at the start of the project followed by INR 9.5 crores at the end of the first month. Thereafter every month end the expenses will reduce by INR 0.5 crores as compared to that incurred in the preceding month till it becomes NIL. It is expected that the rental income from each of the shops will be INR 5 lakhs per month payable in advance starting from 21st month. Maintenance cost of the mall is expected to be INR 10 lakhs per month payable in advance starting from 25th month.

- i) Explain what is meant by
 - a) Discounted Payback Period (DPP) of a project; (2)
 - b) Internal Rate of Return. (2)
 - ii) Calculate the present value of cost of construction at an effective rate of interest of 9% p.a. effective. (5)
 - iii) Calculate the DPP for the project at an effective rate of interest of 9% p.a. effective. (8)
 - iv) After 15 years from the start of construction, the builder decides to sell the shopping mall. Calculate the price at which the mall should be sold to get an IRR of 9% p.a. effective. (4)
- [21]

- Q. 6)**
- i) Explain the difference between the redemption yield and the running yield. (2)
 - ii) An investor made an investment of INR 1,00,000 for the purpose of meeting the liability of INR 2,50,000 which is due after 15 years. Calculate the probability that the investor will be able to meet the liability payout.

Following are the assumptions he made about the future interest rates (i_t):

- $1 + i_t$ is log-normally distributed where it is the yield in year t ;
 - The mean and standard deviation of the yield, i_t on a fund in any year are 0.08 and 0.1 respectively;
 - Yields are independent from year to year.
- (6)
[8]

- Q. 7) i)** Explain the term immunization and state three conditions for Redington's immunization. (3)
- ii)** An investor has to make payments of INR 1,00,000 at the end of each 13th and 17th year. Calculate the present value, duration and convexity of a cashflow sequence. You can assume that interest rates at all durations are currently 8% per annum effective. (5)
- iii)** Calculate the amount of money to be invested in 10-year zero-coupon bond and a 20-year zero-coupon bond such that the present value and duration of the assets will be equal to that of liability. You can assume that interest rates at all durations are currently 8% per annum effective. (5)
- iv)** Comment whether the investor is immune to small changes in interest rates based on the conditions for Redington's immunization. (2)
- [15]

- Q. 8)** Three bonds paying annual coupons in arrears of 5%, 8% and 12% per annum effective, respectively and are redeemable at INR 105 per INR 100 nominal each. Their redemption dates are exactly in one, two and three years' time, respectively. The price of each of the bonds is INR 102 per INR 100 nominal.
- i)** Determine the gross redemption yield of all bonds. (6)
- ii)** Calculate spot yields for year 1, 2 and 3. (5)
- [11]
